Response to Reviewer A

Manuscript 292892: “Who Gets an ESG Score? Coverage Selection and Firm Value in Five Southeast Asian Markets”

Southeast Asian Journal of Economics

Comments are quoted in full in italics, and our responses follow in plain text. Section, table, and equation numbers refer to the revised manuscript, in which every sentence that is new or differs from the submitted version is highlighted in blue; text we deleted is not shown. Some secondary results moved to Appendix A of the supplementary material.

Response to Reviewer A

We thank the reviewer for a careful and constructive report. The reviewer’s main point, that the evidence supports selection into coverage more firmly than a valuation effect of coverage, is correct, and the revision is organized around it. We respond to each comment in turn.

Comment 1: First recorded score and publicly observable coverage

“My main concern is whether the first year in which an LSEG ESG score appears in the database represents a meaningful treatment event. Section 3.2 acknowledges that the recorded fiscal year may precede the actual publication of the score and that historical scores may have been backfilled or subsequently revised.

This distinction is particularly important because the study seeks to estimate the valuation consequences of entering ESG coverage. A score recorded retrospectively in a database is not necessarily information that was available to investors at that time. If the actual publication date differs across firms, the estimated treatment effects may combine pre-publication and post-publication observations.

The authors attempt to address this issue by shifting the treatment year forward by one or two years. While these exercises are useful, they do not establish when investors could first observe the scores.

I encourage the authors to investigate whether historical coverage information or archived score publication dates can be obtained, at least for a subset of firms. If this is not feasible, the paper should distinguish more carefully between the first recorded ESG score and the actual initiation of publicly observable ESG coverage. The interpretation of the treatment effects and the claims made in the abstract and conclusion should reflect this distinction."

We agree, and we have revised the manuscript and the claims throughout. We could not date publication with the data we hold, so we now distinguish the first recorded score from publicly observable coverage in every section and draw conclusions only where they hold across datings.

We looked for the dates. Our extract of LSEG Workspace contains each score by fiscal year, with no publication or update date, and the scores it holds are those in the database at download, after any revision. We found no public archive of the provider's coverage lists for the five markets. LSEG does distribute a point-in-time version of its ESG data that keeps values as first published (LSEG, 2023), which would date publication for the cohorts it covers; our extract does not contain it, and Section 3.2 now says so. [AUTHOR_INPUT_NEEDED: whether the authors can obtain LSEG ESG point-in-time data; if so, the publication-dated analysis should be added before resubmission.]

The revision makes four changes. First, the event is now called the first recorded score, defined as the first fiscal year for which the database holds a score (Sections 1 and 3.1); "first scored in year g" is used as shorthand for a first recorded score in year g. Second, a new Section 3.2, "Event Dating, Outcomes, and Recording Errors," explains why the recorded year can precede publication and states the one property we can establish: publication follows the end of fiscal year g, so the recorded year can date entry too early but, unless the provider deleted an earlier score, not too late. It reads in part: "We therefore keep two concepts apart. The first recorded score is what we observe; publicly observable coverage begins at an unknown date from g + 1 onward. We report the valuation path with coverage dated at g, at g + 1, and at g + 2, and draw conclusions only where they hold across these datings." Third, Section 5.1 notes that the selection finding does not depend on the dating, because the run-up occurs before the first recorded score and therefore before any possible publication. Fourth, the abstract and the conclusion no longer state a post-coverage estimate as a valuation effect. The abstract now says that "whether coverage itself changed valuations depends on when investors first saw the score

and on untestable trend assumptions," and the conclusion says that the answer "moves from no detected change (95% interval −0.046 to 0.048 in log MTB) to a decline of 6.6% depending on when investors are assumed to have seen the score."

We agree that later dating does not establish when investors saw the scores. Its value in the revision is different: Section 4.2 uses it to show what changes when the event moves later, and Comment 3 below describes the result.

Comment 2: Conditional parallel trends

"The validity of the difference-in-differences estimates depends heavily on the conditional parallel trends assumption. However, the empirical evidence raises concerns about whether this assumption is reasonable in the present setting.

Section 5.2 reports that firms entering ESG coverage experienced approximately 10% greater growth in their market-to-book ratios before the recorded treatment year. Although the joint pre-trend test for the primary outcome does not reject at the 5% level (p = 0.074), this result should not be interpreted as evidence supporting parallel trends. The significant pre-treatment differences at individual event years and the results for market capitalization warrant particular attention.

The placebo exercise in Table 3 (R5) also produces a significant positive estimate for market-to-book ratios. I recognize that the authors correctly describe this result as another manifestation of the pre-coverage growth pattern rather than independent evidence. Nevertheless, it reinforces the difficulty of constructing a credible counterfactual for the treated firms.

The sensitivity analyses in Section 5.2 are helpful, but they also show that relatively modest departures from parallel trends can materially change the interpretation of the results. I would like the authors to explain more clearly why conditional parallel trends might hold after the base year despite the observed differences in earlier growth trajectories.

The paper would also benefit from a clearer distinction between the observed evolution of valuation differences and the causal effect attributed to ESG coverage. If the identification assumptions cannot be supported convincingly, the conclusions should place greater emphasis on the descriptive evidence concerning selection into coverage."

We agree with each point and have revised the text accordingly. The pooled sample fails the pre-trend diagnostic, the post-coverage estimates are now presented as conditional on the parallel trends assumption, and the conclusions rest mainly on the descriptive evidence of selection.

On the pre-trend test, Section 4.1 now states: “The MTB non-rejection is not evidence of parallel trends: the estimates for event years −5 and −4 differ from zero, and pre-trend tests have low power (Roth, 2022).” On the placebo, Section 4.3 now states that its failure restates the pre-coverage growth in Figure 1. On the sensitivity analyses, Section 4.3 now concludes that “departures from parallel trends smaller than those observed before coverage are therefore enough to change the reading of either dating.”

On why conditional parallel trends might hold after the base year, a new paragraph in Section 3.3 gives two arguments for and one against. In the Malaysian wave, which carries 44.1% of the weight in the pooled estimate, the pre-coverage estimates show no detectable run-up (joint pre-trend $p = 0.549$), so for these firms the assumption does not require a trend to stop. We say in the same paragraph that this is not evidence of parallel trends: the run-up interval for the wave runs from −7.4% to 15.2%, and the wave split is exploratory. If the provider adds firms once they cross a size or visibility threshold, relative growth stops being a reason for selection once the threshold is crossed, and conditioning on base-year size compares firms on the same side of it. Against these, if part of the run-up was transitory, scored firms would have fallen back without a score, and the estimates would understate any effect. The paragraph ends: “We therefore treat the post-coverage estimates as conditional on the parallel trends assumption and give most weight to evidence that does not depend on it: the ordering of growth and entry, and the contrast between waves with and without a run-up.”

On the distinction between observed evolution and causal effect, the text now separates the two wherever results are reported. Section 4.1, “Who Gets Scored and the Valuation Path Around Entry,” describes the observed gap and adds: “Only reading the post-coverage part as an effect of coverage depends on the parallel trends assumption.” Section 4.2 reports the average change after entry as conditional on parallel trends, and Table 2 is retitled in the same way. The emphasis of the conclusions has moved to selection, as described under Comment 8.

Comment 3: Baseline and one-year-later datings

"The difference between the baseline estimate and the alternative treatment timing specification deserves more attention.

The baseline estimate for log market-to-book is 0.002, whereas shifting the treatment date forward by one year produces an estimate of −0.068 (Table 3, R10). The latter estimate is statistically significant and economically meaningful.

I acknowledge the authors' discussion of possible mean reversion and the changing composition of cohorts across event years. However, the two specifications appear to support rather different interpretations of what happens around the initiation of ESG coverage.

It would be useful to disentangle the effects of changing the treatment date, changing the reference period, and changing the estimation sample. In particular, comparisons using a common set of firms and cohorts would help establish whether the differences arise primarily from treatment timing or sample composition.

The authors should also explain which treatment definition is most defensible from an economic perspective. Given the uncertainty surrounding actual score publication dates, the baseline specification should not automatically receive greater weight simply because it was pre-specified."

We agree and have carried out the decomposition the reviewer suggests. The new Section 4.2 and Table 4 move from the first-recorded-year estimate to the one-year-later estimate one step at a time on a common set of firms, and Table S6 repeats both datings by coverage wave.

Each row of Table 4 changes one feature. Restricting to firms observed in both g − 1 and g, so that every later row draws on the same set of firms, changes the MTB estimate by −0.005. Averaging event years 1 to 4 instead of 0 to 3, with the reference year unchanged, changes it by −0.016 (standard error 0.009). Moving the reference year from g − 1 to g, which is what a one-year-later dating does to the comparison, changes it by −0.044 (standard error 0.019), 62% of the total gap of −0.070. Measuring size in g and returning to all firms change little (0.002 and −0.008). Rows 7 to 10 repeat the steps for the cohorts observed through event year 4 and show the same pattern. The difference between the two estimates therefore comes mainly from the

reference period, not from sample composition. Its economic meaning is that the later dating compares every later valuation with the first recorded year, after each cohort's own rise in that year: moving the reference year subtracts from each later estimate the change of the same scored firms, relative to the same never-scored firms, between g − 1 and g. The step of −0.044 is each cohort's own change between g − 1 and g averaged with the weights of event years 1 to 4, an accounting identity rather than a separate estimate, so part of the decline under the later dating is the rise in the first recorded year itself. Because the steps are sequential, the shares depend on their order; we take the window step before the reference step.

The wave comparison in Table S6 adds the most informative evidence. Under the one-year-later dating the MTB change is −0.136 for the Thai and Singaporean wave and −0.111 for the remaining cohorts, both of which had a run-up, but 0.012 for the Malaysian wave, in which no run-up was detected (2.8%, 95% interval −7.4% to 15.2%); its first-recorded-year estimate is 0.021. Where no run-up was detected before entry, neither dating detects an MTB change (95% intervals −0.053 to 0.091 and −0.055 to 0.076).

On which definition is most defensible, we agree that the baseline should not receive more weight because it was pre-specified, and Section 4.2 now says so: "Pre-specification does not make the recorded year the date on which investors learned of coverage, so we give the two datings equal standing." We report both estimates as answers to different questions. Neither isolates an effect of coverage, because the later dating measures valuations from the first recorded year, after the rise in that year. Section 4.2 states the two readings, a reversal of the run-up that could hide a gain or a negative effect of coverage, and that the data cannot separate them. What the datings share is that neither shows a gain and that the decline appears only where a run-up preceded entry.

Comment 4: Common support and size adjustment

"The paper documents substantial differences between firms that receive ESG scores and those that remain unscored. According to Table 1, scored firms are considerably larger, and Section 4.2 reports that 24.6% of treated firms have base-year total assets exceeding the 95th percentile of the never-scored firms.

This raises concerns about common support and the extent to which the control firms provide a credible counterfactual for the treated firms.

The decision to use outcome regression rather than the doubly robust estimator is understandable given the reported overlap difficulties. However, outcome regression does not eliminate the underlying support problem and may rely heavily on functional-form assumptions when treated firms lie outside the observed range of the controls.

The additional analysis restricting treated firms to the control group's size range is therefore important. I suggest giving this exercise greater prominence in the main text and reporting more information about the characteristics of the firms retained and excluded.

The authors should also clarify how sensitive the results are to the specification of the size adjustment. This is especially relevant because firm size is closely related to both the probability of receiving ESG coverage and the valuation outcomes being examined."

We agree. The size-range restriction is now in the main text with a new part of Section 4.3 on firm size and common support and two new tables. Section 3.3 now states that outcome regression "extrapolates the relation estimated among controls to scored firms larger than most controls."

Table 5 varies the size adjustment and the support. Across functional forms (no covariates, linear, quadratic, cubic, quintile indicators, and size measured at g − 3), the MTB estimate stays between −0.010 and 0.025. Keeping scored firms no larger than the 95th percentile of controls in their cell retains 487 scored firms and gives −0.021 (p = 0.426); the 90th and 99th percentile cut-offs give −0.013 and −0.006. Within the restricted support, the doubly robust estimator (with control propensity scores capped at 0.99) becomes usable and gives −0.010 (p = 0.715), with 1.1% of scored firm-cells having an estimated propensity score above 0.95 (Sant'Anna & Zhao, 2020); the size restriction follows the spirit of the overlap trimming of Crump et al. (2009), applied to size rather than to the propensity score. The MTB result therefore does not depend on extrapolating the size adjustment.

Market capitalization is more sensitive, as the reviewer anticipates: its estimate is 0.016 without covariates, between −0.059 and −0.033 across the functional forms that adjust for size, and

between −0.096 and −0.061 on the restricted supports. Section 4.3 now says that the negative market capitalization change depends on how size is modeled.

Table 6 describes the firms retained and excluded. The 138 excluded firms are large (median base-year total assets of 2,351.7 million U.S. dollars against 170.9 million) and more often Singaporean, Indonesian, or Thai, whereas 71.5% of the retained firms are Malaysian. The text states that the restricted estimate describes mid-sized scored firms, largely from the Malaysian expansion, and says little about the largest firms.

Comment 5: Mechanisms

"The discussion draws on investor recognition and ESG preference theories to motivate the expected valuation effects of ESG coverage. These are reasonable theoretical starting points, but the empirical analysis does not directly observe the mechanisms through which coverage might affect firm value.

In particular, the study does not contain information on institutional ownership, ESG-related fund holdings, trading activity, or changes in investor participation. It is therefore difficult to determine whether the absence of a detectable valuation increase reflects limited investor recognition, weak demand for ESG-rated firms, or other factors.

The high-score versus low-score comparison in Section 5.3 also requires a more cautious interpretation. The insignificant difference between the two groups does not necessarily imply that ESG preferences are unimportant. The test has limited power to detect moderate differences, and the initial ESG score may be correlated with firm characteristics such as size.

I suggest that the authors distinguish more explicitly between theoretical mechanisms that motivate the hypotheses and mechanisms that can actually be evaluated with the available data. The empirical evidence appears better suited to examining selection into ESG coverage and subsequent valuation patterns than to distinguishing between investor recognition and ESG preference channels."

We agree and have revised three places. A new paragraph at the end of Section 2.4 separates the motivating mechanisms from what the data can evaluate: "Without ownership, fund holding, or trading data, we cannot observe the channels themselves; the data show whether entry follows

relative growth, how the valuation gap evolves around entry, and whether that path differs with the initial score." The wording of H3, which the original version described as a test that "separates" the preference channel, now says that a difference would point toward that channel and that the absence of one cannot rule it out. A new Section 5.2, "Mechanisms That the Data Can and Cannot Evaluate," applies the same distinction to the discussion.

The interpretation of the high- and low-score comparison (now Section 4.4) is more cautious. It reports the minimum detectable differences of 0.113 and 0.111, about 12% in the MTB ratio, notes that the score is correlated with size (0.35) and that the residual split removes only part of this link, and concludes that neither split "can detect a moderate difference, so the result does not show that ESG preferences are unimportant."

Comment 6: Missing market-to-book ratios and implausible market capitalization

"Section 3.2 reports substantial differences in the availability of market-to-book ratios between scored and never-scored firms. The supplementary material further indicates that approximately one-quarter of firm-year observations are missing for the primary outcome.

This is potentially important because missing market-to-book observations may be systematically associated with financial distress, negative book equity, firm size, or the likelihood of receiving ESG coverage.

The authors should provide a more detailed assessment of whether the missing observations affect the composition of the treatment and comparison groups. A comparison of firms with and without usable market-to-book data would be informative.

I am also concerned by the acknowledgment in Section 7 that some market capitalization observations are implausibly large. Winsorization may reduce the influence of extreme observations, but it does not necessarily correct underlying data errors. The authors should explain how these observations were identified and whether the main results are robust to excluding potentially erroneous records."

We agree that missing ratios may be systematic, and our checks confirm it. A new Section 4.3, new Table 7, and new Table S8 address both concerns. In checking this comment we also

corrected a statement in the original Section 3.1: the source data contain no MTB ratio at or below zero, so firm-years with negative book equity appear as missing together with firm-years that lack market or accounting data, and we cannot separate the two. The revised Section 3.1 says so.

All missing-data shares are now computed over listed firm-years: every firm has a row for each year from 2014 to 2024, and the 3,647 rows before a firm lists carry no data. On this basis, MTB ratios are missing for 18.5% of never-scored and 9.0% of scored firm-years. Table S8 compares firm-years with and without an MTB ratio. Firm-years without one are more often highly levered (leverage above 0.5 in 25.4% of never-scored firm-years without a ratio, against 8.4% with one) and more often Indonesian (36.2% against 24.1%), consistent with financial distress and with gaps in market data. Only 36.7% of never-scored and 36.6% of scored firm-years without a ratio have a market capitalization, so most gaps are missing market data rather than negative book equity alone. Of the 50 scored firms that do not enter the MTB estimate, those with base-year total assets have a median similar to that of firms that do (280.3 against 294.3 million U.S. dollars), so the missing ratios do not shift the median size of scored firms.

Table 7 tests whether the gaps change the estimates. Keeping only firms with a complete MTB series gives 0.008, and keeping only scored firms with complete series gives 0.006, against the baseline 0.002. Estimating market capitalization only on firm-years that have an MTB ratio moves its change from −0.055 to −0.023, so part of the negative market capitalization estimate comes from firm-years that drop out of the MTB sample; Section 4.3 reports this.

On the implausible market capitalization records, Section 3.2 now explains the rule we use to identify them. Market capitalization divided by the MTB ratio gives the implied book equity, which cannot exceed total assets. Because ratios slightly above one can arise from different measurement dates, we flag firm-years in which the implied book equity exceeds ten times total assets, an error of an order of magnitude. The rule flags 209 of 27,127 firm-years (0.77%), 73 of them for scored firms; the largest flagged value is an Indonesian firm recorded at 1,138.8 billion U.S. dollars. Removing the flagged firm-years gives −0.007 for MTB and −0.045 for market capitalization. Removing every firm with a flagged year, flagging at twice instead of ten times total assets, or winsorizing within each market and year gives MTB estimates between −0.019

and 0.005. None of these checks produces an MTB gain. Section 5.3 now describes the remaining limitation.

In checking the data for this comment, we also found that every firm in our extract has data in 2024: firms enter the panel when they list, but firms that delisted between 2014 and 2023 are not in the extract. Section 3.1 now states that the sample conditions on survival to 2024, Section 3.2 notes that the absence of any firm losing its score fits a download of the provider's current universe, and Section 5.3 lists survivorship as a limitation whose bias we cannot sign without delisting data.

Comment 7: Uneven distribution across markets

"Although the study covers five Southeast Asian markets, the distribution of treated firms is highly uneven. Malaysia and Thailand account for the overwhelming majority of firms entering ESG coverage, while the samples for Indonesia and the Philippines are particularly small.

The supplementary analyses are useful in showing how the estimates vary across countries. However, the limited number of treated firms in some markets makes it difficult to draw meaningful country-specific conclusions.

The Malaysian coverage wave also differs considerably from the other cohorts in terms of pre-treatment valuation dynamics. This raises the question of whether the pooled estimate represents a reasonably coherent treatment effect or averages across substantially different coverage processes.

I suggest that the authors discuss more explicitly what the pooled estimates represent, particularly given the concentration of treatment observations in a few markets and years. The interpretation should acknowledge that the results primarily reflect the markets and coverage waves with the greatest representation in the sample."

We agree. A new part of Section 4.4 on what the pooled estimate represents and a new Table S10 report the weight of each market, wave, and year. Malaysian firms carry 55.2% of the weight in the headline MTB estimate and Thai firms 26.7%; the Malaysian wave alone carries 44.1%, and firm-years from 2021 to 2024 carry 87.2%. The section states that "the pooled estimate therefore describes mainly the Malaysian expansion and the Thai cohorts after the COVID-19 pandemic,

not an average effect for the five markets," and that the country estimates for Indonesia, the Philippines, and Singapore cannot support conclusions for those markets.

On whether the pooled estimate averages different coverage processes, the revision treats the waves as distinct throughout. Section 4.1 reports the paths by wave, and Section 4.2 and Table S6 show that the waves differ under the later dating as well: the Malaysian wave, in which no run-up was detected, shows no detectable change under either dating, while the waves with a run-up show the decline. We now use this contrast as evidence (Sections 3.3 and 4.2) rather than averaging over it. Section 4.2 now also tests the differences between waves with joint firm-bootstrap draws: relative to the Malaysian wave, the Thai and Singaporean wave differs in the run-up (-0.258, $p = 0.010$) and in the one-year-later change (-0.148, $p = 0.006$), while the remaining cohorts differ in the one-year-later change but not detectably in the run-up ($p = 0.307$). Because we drew the wave boundaries after seeing the coverage counts, the manuscript labels this comparison exploratory.

Comment 8: Framing of the contribution

"The most convincing contribution of this paper, in my view, is the evidence that entry into ESG rating coverage is selective and may follow substantial changes in firm characteristics and valuation. This is an important issue for empirical studies that compare ESG-rated and unrated firms.

The evidence regarding the valuation effect of coverage itself is less conclusive. The baseline estimates are close to zero, but the alternative timing specifications, pre-treatment dynamics, and sensitivity analyses leave room for different interpretations.

I therefore encourage the authors to reconsider how the paper's contribution is framed. Rather than presenting the absence of a detectable valuation gain as the central finding, the study could place greater emphasis on the empirical challenges created by selective and potentially retrospective ESG coverage.

The implications for investors, rating providers, and regulators should also be reconsidered in this light. In particular, the results do not establish that expanding ESG coverage has no valuation benefits. They indicate that such benefits cannot be reliably identified from the present data and research design without additional assumptions."

We agree and have reframed the paper. Selection into coverage is now the central finding, and the valuation effect is presented as something the data cannot identify without further assumptions.

The abstract now leads with selection and states that “whether coverage itself changed valuations depends on when investors first saw the score and on untestable trend assumptions.” The introduction presents the evidence in this order: entry is selective; after entry the gap stops widening but the data do not pin down a causal effect; and a static regression attributes selection to the score. The contributions are restated as evidence of selection, a demonstration of why the valuation effect is hard to identify, and a warning for empirical work. Section 5.1, “What the Evidence Shows,” now makes the reviewer’s point: “With annual data, a provider that chooses whom to score, and recorded scores that may precede publication, a valuation effect of coverage cannot be reliably identified without additional assumptions.” It also reports that one of many exploratory subgroups, the 58 firms first scored by 2019, shows a positive estimate that we do not read as evidence of a gain. The title already names coverage selection, so we have kept it.

We have rewritten the implications in Section 5.3. For researchers, studies of ESG ratings in the region should model when firms enter coverage, report pre-entry dynamics, and record the download date of their rating data. For rating providers and data users, publishing the date on which each firm’s coverage began, and keeping the vintages of revised scores, would separate what was known from what was filled in later. For exchanges, regulators, and listed firms, the section now states: “Our results do not show that expanding ESG coverage has no valuation benefit; they show that such a benefit cannot be established from these data, so arguments for expanding coverage should rest on grounds other than an expected rise in issuer valuations.” The conclusion has been rewritten in the same terms and ends with the data that would move the question forward: point-in-time records of when each firm’s score was first published, and ownership and trading data.

Summary of changes

Comment	Change	Location in the revised manuscript
A1	First recorded score	Sections 1, 3.1, 3.2, 4.2, 5.1,

Comment	Change	Location in the revised manuscript
	distinguished from publicly observable coverage; datings g, g + 1, g + 2	and 5.4; Abstract
A2	Pre-trend reading corrected; arguments for and against the parallel trends assumption; observed path separated from causal reading	Sections 3.3, 4.1, 4.2, and 4.3; Table 2 title
A3	Step-by-step reconciliation of the two datings on a common sample; datings by wave	Section 4.2; Table 4; Table S6
A4	Size adjustment, support restrictions, doubly robust estimator; retained and excluded firms	Sections 3.3, 4.3; Tables 5, 6
A5	Motivating and testable mechanisms separated; cautious reading of the score split	Sections 2.4, 4.4, 5.2
A6	Source of missing MTB corrected; firm-years with and without MTB compared; recording errors identified and removed	Sections 3.1, 3.2, 4.3, 5.3; Table 7; Table S8
A7	Weight of each market, wave, and year in the pooled estimate	Section 4.4; Table S10
A8	Contribution reframed around	Abstract; Sections 1, 5.1, 5.3,

Comment	Change	Location in the revised manuscript
	selection; implications rewritten	5.4

New material: Tables 4 to 7 in the manuscript; Tables S6, S8, and S10 and Appendix A.8 to A.10 in the supplementary material; the R code that produces them is part of the replication package.

We thank the reviewer again for comments that improved the paper.

Sincerely,

The authors